

Data Engineer

Summary:

- ⇒ In depth knowledge in **Big Data**, Data Engineering, and **Data Warehousing**, including Business Requirements Analysis, Application Design, Data Modelling, Development, Testing, Visualization, and Documentation.
- ⇒ Hands-on **cloud experience** with **Azure**, **AWS**, and **GCP**, specializing in scalable and efficient ETL solutions.
- ⇒ Proven track record in data quality management, data mapping, and using **ETL tools** like Glue, Striim, Matillion, Talend, Informatica, and DataStage.
- ⇒ Expertise in Agile methodologies, delivering high-quality results in collaborative, fast-paced environments.
- ⇒ Advanced proficiency in SQL, with extensive experience in querying and optimizing relational databases (MySQL, PostgreSQL, SQL Server, and Redshift) to support data engineering tasks, ensuring high-performance data retrieval, transformation, and reporting.
- ⇒ Implemented **enterprise data governance** strategies by defining metadata management, data quality, lineage tracking, and access controls in **Snowflake**, **Azure**, and **GCP**, ensuring compliance with industry standards.

Education:

- ⇒ **Lewis University, Romeoville, IL:** Master of Science | **Major:** Data Science | (Jan 2021 - Dec 2022)
- ⇒ **Osmania University, India:** Bachelor of Science | **Major:** Computer Science | (Aug 2016 – Oct 2020)

Technical Skills:

- **Programming Languages:** Python, PySpark, Scala, Java, SQL, R
- **Databases:** SQL (Oracle, MySQL, SQL Server) | NoSQL (MongoDB, DynamoDB)
- **Version Control Tool:** Git, GitHub, Jira, SVN, Azure Repos
- **Cloud Platforms:** AWS (S3, Redshift, RDS, Glue, EC2, EMR, Athena, Kinesis, CloudWatch, CloudTrail, IAM, EC2, EBS) | GCP (BigQuery, Pub/Sub, Google Composer, Google Cloud IAM) | Azure (Synapse, Data Factory, Data Lake, Databricks)
- **Data Visualization:** Tableau, PowerBI, QuickSight
- **ETL Tools:** Talend, Informatica, DataStage, Striim, Glue, Matillion
- **DevOps/Automation:** Jenkins, Terraform, Airflow, Kubernetes, CI/CD pipelines, Azure Pipelines
- **Others:** Putty, Eclipse, IntelliJ, PyCharm, Visual Studio, Linux (Redhat, CentOS), Windows, Mac, Unix Bash, Advanced Excel, Powerpoint, Word, Jupyter Notebook
- **Interpersonal Skills:** Strong Internet Research, Excellent Verbal, Written, and Quantitative Skills, Time Management and Prioritization Abilities, Effective Presentation, Team Player, Strong communication skills, Leadership and Negotiation Skills.

Professional Experience

Data Engineer | TechDemocracy | Ascension Healthcare, Houston, TX

Nov 2023 to present

- Successfully led the migration of **14+ trillion** records across 219 tables for a **Cerner** ministry to Google Cloud Platform (GCP), ensuring **99.996%** data accuracy through meticulous validation and quality assurance processes.
- Designed and implemented a robust batch processing framework using **Python** to develop and manage DAGs executing every 4 hours, seamlessly integrating an **Informix** database as the source and **BigQuery** as the target dataset, optimizing data ingestion, transformation, and loading processes.
- Worked extensively with BigQuery, developing optimized **SQL** queries, partitioned tables, and materialized views to enhance analytics and reporting performance.
- Played a key role in the creation of the **largest data lake** in **North America** using **Striim**, enabling efficient data ingestion, storage, and processing across the enterprise.
- Successfully migrated **batch** processing workflows to Google **Composer** and **Airflow DAGs**, enhancing automation, scheduling, and monitoring capabilities.
- Designed and managed **Pub/Sub** topics, **service accounts**, and **datasets**, provisioning them through **terraform** to streamline data communication and access control within cloud environments.
- Developed and enforced data governance frameworks, ensuring high data quality, regulatory compliance, and security best practices across ETL pipelines and data storage solutions.
- Developed and deployed data engineering solutions for the healthcare industry, ensuring compliance with industry regulations (e.g., HIPAA). Implemented robust data quality frameworks, including automated data cleansing and validation, to maintain data integrity and governance.
- Hands-on experience in Google Cloud Platform (GCP), including managing data pipelines using Google Cloud Storage, BigQuery, Dataform, and Pub/Sub for data ingestion, processing, and analytics; contributing to the development of scalable cloud data architectures.
- Optimized SQL queries, stored procedures, and indexing strategies in SQL Server Management Studio (SSMS) to improve data retrieval speeds and enhance backend system performance for large-scale datasets.
- Leveraged Terraform to automate the creation of **Jenkins** pipelines, accelerating **CI/CD** processes and minimizing human errors.
- Worked extensively with **Kubernetes** for container orchestration, ensuring high availability and scalability of applications.
- Demonstrated proficiency in **SQL Server**, DBeaver, **MongoDB**, **Informix**, and **Oracle SQL** Developer for efficient database management and development.
- Drove **cost optimization** initiatives for applications, achieving significant reductions in operational expenses.
- Developed Change Data Capture (**CDC**) applications using Striim to redirect data to BigQuery, providing faster and more reliable storage solutions for downstream **stakeholders**.
- Collaborated with **DevOps**, Cloud **DataOps**, and other cross-functional teams to identify, fix, and refactor bugs in data pipelines and cloud infrastructure, ensuring smooth operations and continuous improvement across the development lifecycle.

Environment: GCP-Big Query, Cloud storage, Pub/Sub, composer, Airflow, MongoDB, SQL Server, Oracle SQL, Kubernetes, Python, Striim, CI/CD, Terraform

Data Engineer | Devoir Software Solutions, Columbus, OH

Jan 2023 to Oct 2023

- Designed and implemented end-to-end data pipelines using Azure Data Factory, Databricks, and Synapse Analytics, optimizing ETL workflows for performance and cost efficiency.

- Migrated SSIS-based ETL pipelines to Azure Data Factory (ADF) and Snowflake, enhancing scalability and reducing runtime.
 - Developed real-time data pipelines in Snowflake and SQL Server, integrating with Power BI for interactive dashboards.
 - Implemented data modeling best practices (Kimball, DVM) to improve reporting efficiency in Power BI and Cognos.
 - Optimized Power BI datasets by restructuring data models and indexing strategies in Redshift/PostgreSQL, reducing report load times by 40% and enhancing data visualization efficiency.
 - Designed and deployed serverless data processing workflows using AWS Lambda and Step Functions, automating data ingestion from S3 to Redshift and improving pipeline execution time by 25%.
 - Defined best practices for **data modeling**, **lakehouse** architecture, and metadata management, enabling scalable and maintainable data solutions.
 - Developed scalable data models and optimized data lake architectures on **Azure Data Lake** Storage (ADLS) to support structured and unstructured data.
 - Automated data ingestion, transformation, and orchestration using Azure Data Factory, **Apache Spark**, and **Delta Lake**, improving data availability and reliability.
 - Implemented real-time streaming solutions using Azure Event Hubs, Azure Stream Analytics, and Databricks Structured Streaming for low-latency data processing.
 - Contributed to the development of **gun detection** systems, ensuring high accuracy and reliability.
 - Established CI/CD pipelines for data pipelines and **Databricks** notebooks using Azure **DevOps** for version control and automated deployments.
 - Increased system scalability by **25%** through optimized **CI/CD** workflows using **GitHub Actions**.
 - Built and deployed machine learning models on Databricks ML with Azure Machine Learning, supporting advanced analytics and AI-driven insights.
 - Designed and deployed real-time **object detection** models (YOLOv5, YOLOv8) with **Triton**, achieving a 15% improvement in inference speed.
 - Led the development and optimization of ETL pipelines, utilizing **Alteryx** to automate data transformation workflows, ensuring high-quality data wrangling and accurate reporting for operational risk loss measurement. Collaborated with cross-functional teams to integrate Alteryx workflows into cloud-based data platforms AWS, enhancing data transformation and visualization processes.
 - Collaborated with cross-functional teams to define data **architecture standards**, governance policies, and best practices for enterprise-scale deployments.
 - Developed CI/CD pipelines for Databricks notebooks and ADF pipelines using Azure DevOps, GitHub Actions, and Terraform.
 - Ensured data governance, security, and compliance by implementing RBAC, encryption, and access policies across **Azure data services**.
 - Developed and maintained robust ETL pipelines using Python, integrating data from multiple sources via REST APIs, and automating data workflows with AWS Glue, Python scripts, and Apache Airflow to streamline data transformation and loading into cloud-based warehouses like Redshift and Snowflake.
 - Configured Horizontal Pod Autoscaling (HPA) to optimize application performance in **Kubernetes** clusters.
 - Utilized **PostgreSQL** as the primary database for application development, ensuring efficient and secure data management.
 - Developed and hosted **APIs** using **FastAPI**, including GET and POST endpoints, ensuring high-performance and secure API operations.
 - Explored and fine-tuned **LLMs** and **Gen AI** models for data processing, integrating AI-driven insights into data transformation workflows.
 - Developed and maintained automated Power BI dashboards, integrating Snowflake and Excel data to track revenue trends, customer insights, and operational KPIs, reducing manual reporting efforts by 40%.
 - Developed and optimized ETL pipelines using AWS Glue, Python, and PySpark, improving data transformation efficiency by 30% and ensuring seamless integration with AWS S3 and Redshift for large-scale analytics.
 - Implemented and managed infrastructure as code (IaC) using Terraform to automate AWS resource provisioning, reducing deployment time by 40% and ensuring infrastructure consistency for data pipelines.
 - Built and tested SQL queries in Snowflake to extract, transform, and validate large datasets, ensuring high-performance analytics and seamless integration with Power BI dashboards for business intelligence reporting.
 - Built and maintained real-time streaming data pipelines using **Kafka**, optimizing producer-consumer architecture for high-throughput and low-latency data processing.
 - Integrated **Power BI** with **SQL databases**, **AWS Redshift**, and **Snowflake**, enabling seamless real-time reporting and data-driven decisionmaking.
 - Designed and optimized data models in Snowflake to support CRM and ERP reporting needs, improving query performance and data retrieval efficiency for business analytics.
 - Responsible for preparing data lineage documents for the SSIS (SQL Server Integrated services) packages before ETL Re-engineering.
 - Developed and optimized **ETL workflows** using **AWS Glue**, **Python**, and **PySpark**, improving data pipeline efficiency and scalability.
 - Designed and developed ETL workflows in **IBM DataStage**, integrating structured data from multiple sources, optimizing transformation logic, and ensuring efficient data loading into AWS **Redshift** for analytics and reporting.
 - Designed and developed **AI/ML**-specific data pipelines, ensuring seamless ingestion, transformation, and storage of high-volume datasets for machine learning model training and inference.
 - Developed robust data governance policies, ensuring data compliance, security, and risk mitigation for AI-driven applications, aligning with GDPR and HIPAA standards.
 - Administered AWS **Redshift clusters**, optimizing performance, workload management (WLM), and security policies to ensure efficient query execution and cost-effective scaling.
 - Integrated data from **REST APIs** using Python, automating data extraction and transformation processes for seamless ingestion into cloud-based data warehouses.
 - Built feature engineering pipelines leveraging Python, Apache Spark, and Snowflake, optimizing data preprocessing for Artificial Learning/ Machine Learning training workflows, reducing model training times by 40%.
 - Migrated data from **Teradata to Amazon Redshift**, leveraging AWS Schema Conversion Tool (SCT) for seamless schema transformation.
 - Integrated **Terraform** into **CI/CD pipelines** in collaboration with DevOps, ensuring automated and consistent infrastructure provisioning.
 - Managed and transformed data in **Amazon EMR** by rewriting **SSIS packages** using **Spark SQL** and **PySpark** for enhanced performance.
 - Automated ETL batch processing with **Apache Airflow**, orchestrating workflows for data ingestion into **Snowflake** for analytical processing.
- Environment:** AWS- S3, Redshift, Glue, Kinesis, Lambda, EMR, Python, PySpark, ETL, Snowflake, SSIS, Spark SQL, Airflow, CI/CD, Terraform, Power BI, Data Pipeline

Data Engineer | We-Do-IT, Hyderabad, India

Aug 2018 - Nov 2020

- Worked with **Azure Data Factory**, **Azure Synapse**, and **Snowpark**, optimizing ETL workflows and data transformations for high-performance analytics and seamless data integration.
- Developed **ETL** Processes in **Databricks** to extract data from **Redshift**, perform transformations and load data to **S3**- Datalayer in Databricks.
- Integrated **Docker** and **Kubernetes** for containerization and orchestration of batch and real-time pipelines, while leveraging **CI/CD** tools like **Jenkins** for automated deployment, reducing manual intervention and improving **data validation** by **20%**.
- Improved query performance and cost efficiency by tuning **Databricks** clusters, optimizing Spark jobs, and implementing Delta Lake optimizations (Z-ordering, Optimize, Vacuum).
- Designed and deployed enterprise data architectures on Azure, integrating Databricks, Synapse, and Power BI for advanced analytics and reporting.
- Designed microservices-based data processing architectures with Azure Kubernetes Service (AKS) and Azure Functions.

- Designed and implemented batch data pipelines using **Python** frameworks like **PySpark** and **Pandas**, automating **ETL** workflows with **Apache Airflow**.
- Automated **ETL workflows** using **AWS Glue, Lambda, and Step Functions**, streamlining data ingestion from diverse sources into **Snowflake and Redshift**.
- Oversaw both **PaaS** and **IaaS** environments, ensuring seamless integration and scalability.
- Managed and governed Reverse ETL workflows, ensuring efficient and secure data synchronization from Snowflake to source systems for operational and analytical use cases.
- Designed and implemented middleware solutions for real-time and batch data processing, integrating multiple enterprise applications using APIs and cloud-native services.
- Developed and deployed infrastructure automation scripts with Pulumi, **optimizing cloud-based data workflows** for operational data processing **and enhancing system reliability**.
- **Led** the migration of on-premises Hadoop/Spark clusters to Azure Databricks, reducing infrastructure costs by 60%.
- Built scalable data solutions for manufacturing analytics, **leveraging** streaming data frameworks **to improve production efficiency and monitor IoT sensor data in real time**.
- Worked on AWS **Data Pipeline** to configure data loads from **S3** to into **Redshift**.
- Developed SQL queries and optimized SAP HANA data retrieval, improving performance for real-time supply chain metrics dashboards.
- Designed and implemented ETL pipelines for SAP ECC and HANA datasets, extracting and transforming large-scale supply chain data for analytics and reporting.
- Developed optimized queries and data models in TIBCO Data Virtualization, improving query response times for real-time reporting applications.
- Integrated TIBCO with AWS S3 and PostgreSQL, enabling seamless data federation across enterprise data sources.
- Designed and implemented data ingestion from **Kinesis Data Streams** (KDS) to **Elasticsearch**, enhancing search capabilities and real-time analytics for large-scale datasets.
- Used **JSON schema** to define table and column mapping from S3 data to Redshift.
- Worked on **Snowflake** Schema, **Data Modeling** and Elements, and Source to Target Mappings, Interface Matrix and Design elements.
- Experience in loading data into **Snowflake DB** in the cloud from various sources and validating the data feed from the source system to **Snowflake DW** Cloud platform.
- Designed and implemented **Kafka**-based data pipelines that processed terabytes of data per day.
- Developed and maintained **Python**-based **ETL** scripts for processing and transforming large datasets, ensuring efficient data pipelines for analytics and reporting.
- Developed **Power BI dashboards** by integrating data from **AWS Redshift, Snowflake, and SQL databases**, providing interactive visualizations and real-time business insights.
- Designed and implemented **ETL pipelines** to extract data from **Oracle** and load it into **Snowflake**, leveraging **Matillion** for data transformation and orchestration.
- Designed and implemented a fully automated **ETL pipeline in AWS** using **Python** and Apache **Airflow** on Amazon **MWAA** (Managed Workflows for Apache Airflow), orchestrating batch data processing every 24 hours.
- Developed and optimized **complex SQL queries** for efficient data extraction, transformation, and loading (ETL) in Amazon **Redshift**, improving query performance and reducing execution time.
- Developed and deployed **RESTful APIs** and microservices using **Python & Flask**, integrating with **Kafka** and **Snowflake** for real-time data ingestion and transformation across distributed systems.
- Partnered with cross-functional **stakeholders**, translating business needs into actionable data strategies and ensuring seamless communication between engineering and analytics teams.
- Developed high-performance data processing frameworks leveraging Flink and Kafka, enabling real-time streaming analytics for event-driven architectures.
- Implemented **DBT** (Data Build Tool) transformations in **Snowflake**, improving data modeling, pipeline automation, and analytics performance.
- Developed and maintained DBT (Data Build Tool) models, implementing DBT tests to ensure data integrity, quality, and transformation consistency in data pipelines.
- Designed and maintained dbt models, tests, and job runs for Snowflake transformations.
- Developed enterprise-wide data architecture, aligning business goals with scalable cloud solutions, ensuring optimized data governance and security.
- Integrated various data sources into **Tableau**, including relational databases, data warehouses, and cloud-based storage, enabling seamless data connectivity for real-time analytics and reporting.
- Worked on Data migration from On-premise Microsoft SQL Server to Amazon Redshift and ETL re-engineering of **SSIS** packages to Spark SQL/python code using AWS EMR.
Environment: AWS- S3, Redshift, Glue, Kinesis, Lambda, Step Functions, `EMR, Python, PySpark, ETL, Snowflake, SSIS, Spark SQL, Airflow, CI/CD, Terraform, Power BI, Data Pipeline, Oracle, SSIS, Tableau, PowerBI, Matillion, MWAA, SQL, Kafka, Azure Data Factory, Azure Synapse, Snowpark

Data Engineer Intern | We-Do-IT, Hyderabad, India

Jan 2018 - Aug 2018

- Assisted in the design, deployment, and management of data infrastructure on **AWS** (Amazon S3, Redshift, AWS Glue), helping automate data pipelines using **AWS Data Pipeline** and **Python** scripts.
- Supported the development of interactive **data visualizations** and **dashboards** using **AWS QuickSight** and **Tableau**, contributing to realtime insights for stakeholders.
- Assisted in integrating **Tableau** with **Amazon Redshift** and **Amazon Athena**, helping the team enable smooth data exploration and analytical reporting from cloud-based data sources.
- Assisted in tuning **Informatica mappings, sessions, and workflows**, optimizing performance and ensuring efficient data processing across various systems.
- Supported the integration of data from **on-premises** and **cloud-based systems** using **Informatica PowerCenter** and **Talend**, ensuring seamless data flow and synchronization.
- Gained hands-on experience with **Hadoop** components (**HDFS, Hive, Pig**) and big data processing tools (**MapReduce, Spark**), performing large-scale data storage, processing, and analysis tasks.
- Contributed to **Hadoop High Availability** setup by configuring **Zookeeper** and working with failover controllers, ensuring a scalable and fault-tolerant data processing environment.
Environment: AWS- Quicksight, Redshift, Athena, S3, Glue, HDFS, Hive, Pig, Hadoop, MapReduce, Spark, Informatica, Tableau, Talend, Python